Dimension Reduction Techniques

17 January 2026 17:39

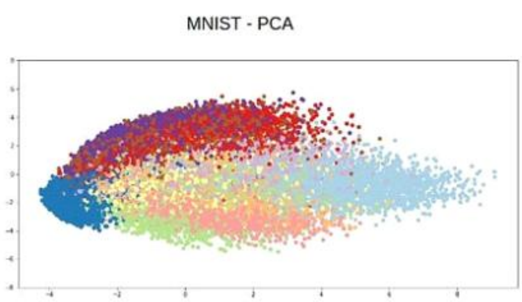
Linear

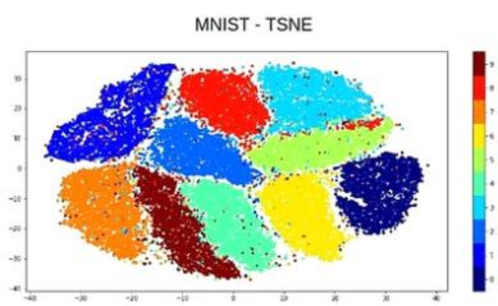
ICA
SVD
NMF

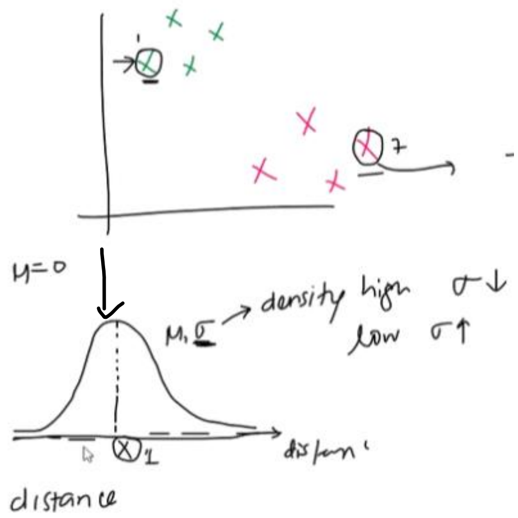
RFE
L_1
Chi-sq (Categorical)
RF

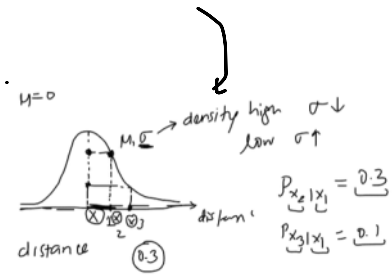
Nonlinear

t-SNE
UMAP
DL → Autoencoder





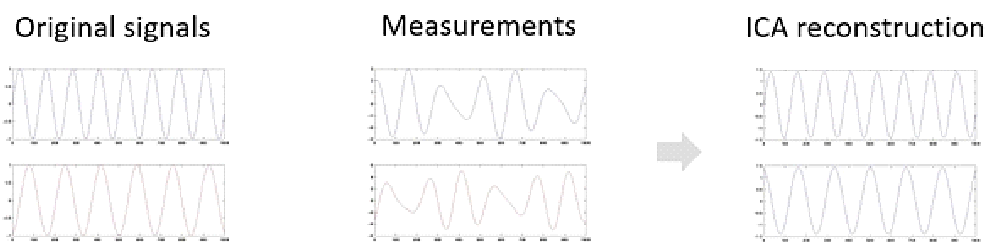


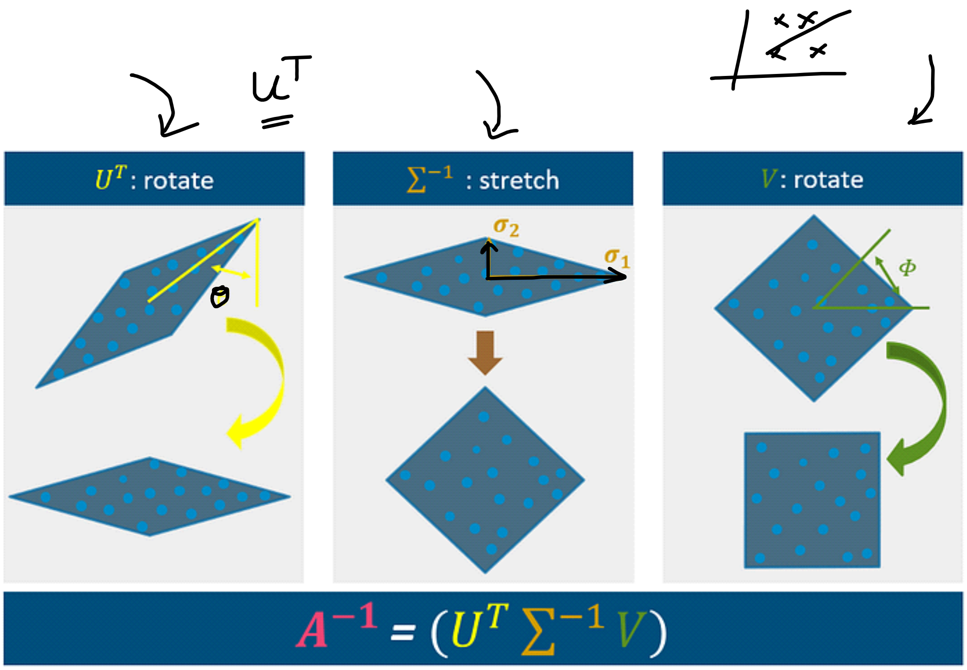


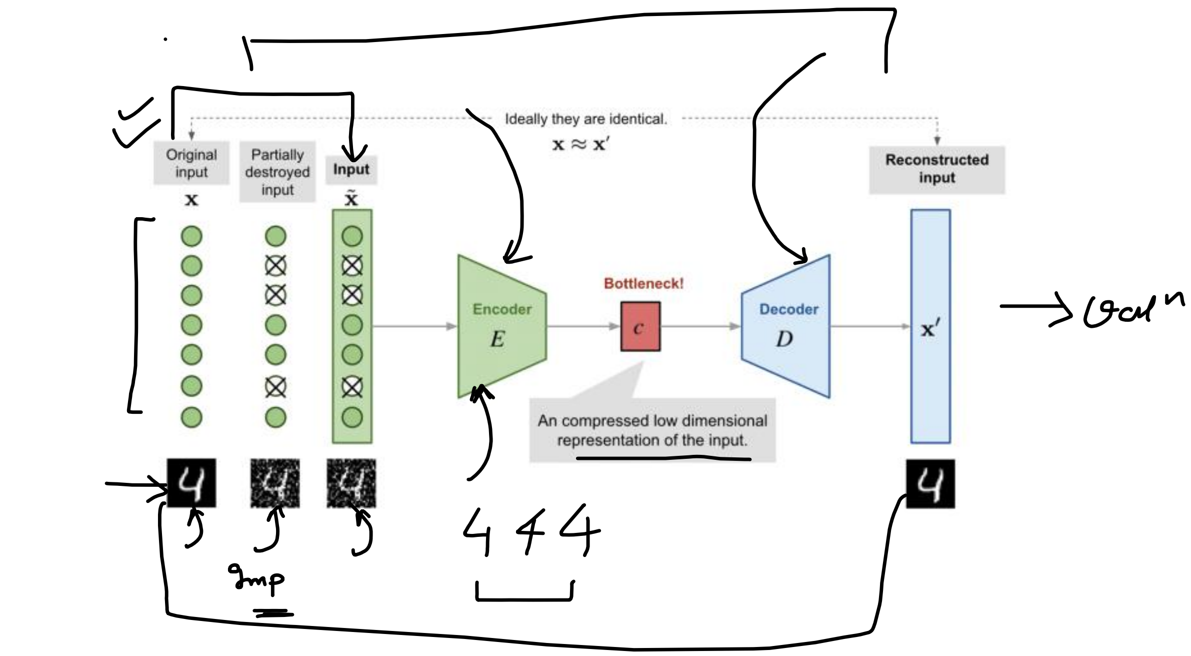
$$p_{j|i} = \frac{\exp\left(-\|x_i - x_j\|^2 / 2\sigma_i^2\right)}{\sum \exp\left(-\|x_i - x_j\|^2 / 2\sigma_k^2\right)}$$

Algorithm 1: Simple version of t-Distributed Stochastic Neighbor Embedding.

Data: data set $\mathcal{X} = \{x_1, x_2, ..., x_n\}$,
cost function parameters: perplexity $Perp$,
optimization parameters: number of iterations T, learning rate η, momentum $\alpha(t)$.
Result: low-dimensional data representation $\mathcal{Y}^{(T)} = \{y_1, y_2, ..., y_n\}$.
begin
  compute pairwise affinities $p_{j|i}$ with perplexity $Perp$ (using Equation 1)
  set $p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}$
  sample initial solution $\mathcal{Y}^{(0)} = \{y_1, y_2, ..., y_n\}$ from $\mathcal{N}(0, 10^{-4}I)$
  for *t=1* **to** T **do**
    compute low-dimensional affinities q_{ij} (using Equation 4)
    compute gradient $\frac{\delta C}{\delta \mathcal{Y}}$ (using Equation 5)
    set $\mathcal{Y}^{(t)} = \mathcal{Y}^{(t-1)} + \eta \frac{\delta C}{\delta \mathcal{Y}} + \alpha(t) \left(\mathcal{Y}^{(t-1)} - \mathcal{Y}^{(t-2)}\right)$
  end
end







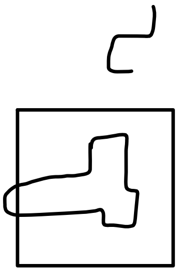


Image →

$C_1 \rightarrow I_{20}$

$C_2 \rightarrow I_{20}$

$C_3 \rightarrow I_{26}$

$\vdots$

$C_n \rightarrow I_{20}$

compression ×

$C_1, C_2, C_3 \rightarrow \boxed{\text{CNN}} \rightarrow C_1, C_2, C_3 \rightarrow$

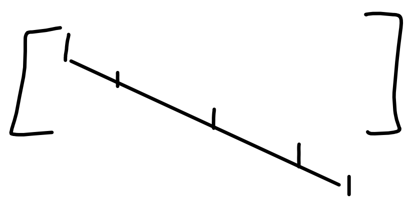
∂prop $\rightarrow A = \underline{U \Sigma} V^T$

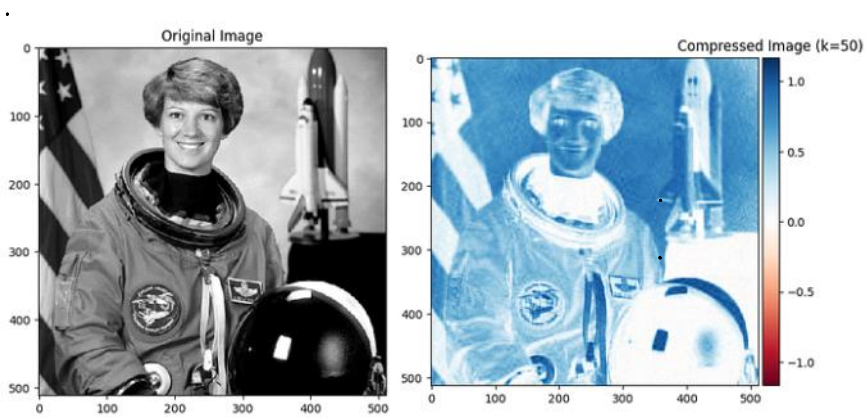
$A \in R^{m \times n}$

$U \in R^{m \times m}$

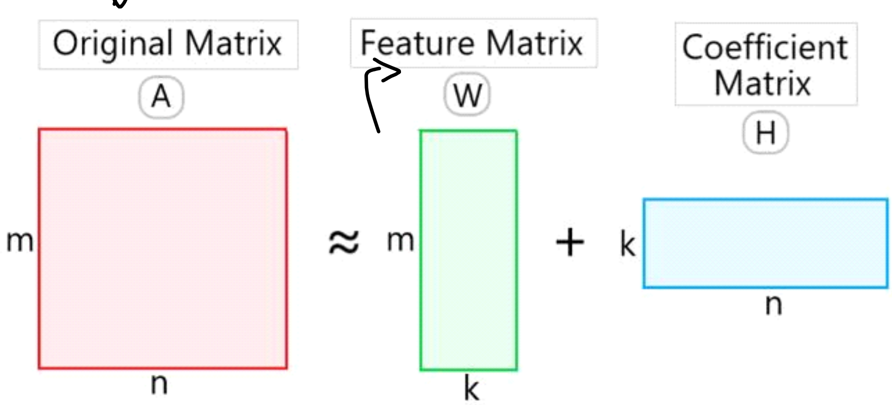
$\Sigma \in R^{m \times n}$

$V \in R^{n \times n}$




Original Image
Compressed Image (k=50)

$X \approx WH$


Original Matrix
A
Feature Matrix
W
Coefficient Matrix
H
m
n
≈
m
k
+
k
n